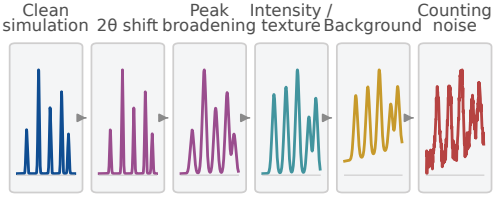
Fig. 6 | Robustness to synthetic experimental artefacts

"Noisy" = synthetic perturbations of simulated inputs; not real experimental measurements.
All results: N=500/config.

Clean/noisy ladders compare clean-trained + clean input with noise-finetuned + noisy input; percentages are match rates.

a Synthetic experimental artefacts

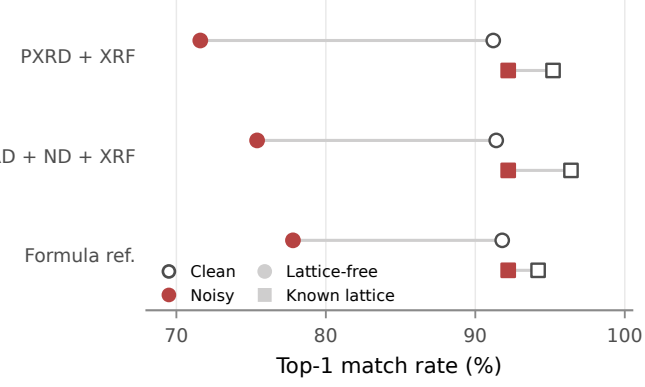
Online perturbation workflow • schematic, qualitative, not to scale



NOISY = synthetic perturbations of simulated inputs

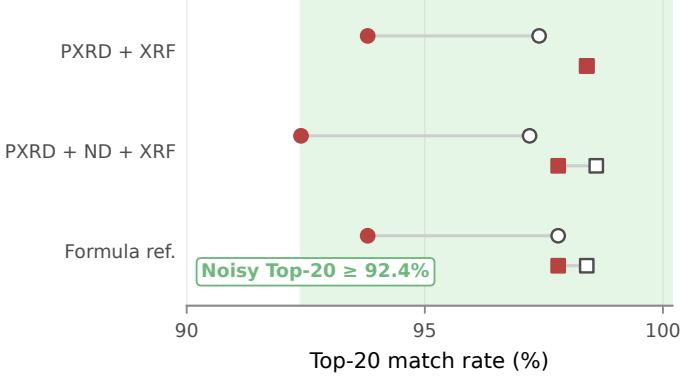
b Top-1 under high-information inputs

Clean ○ / noisy ●; circle free, square known; N=500



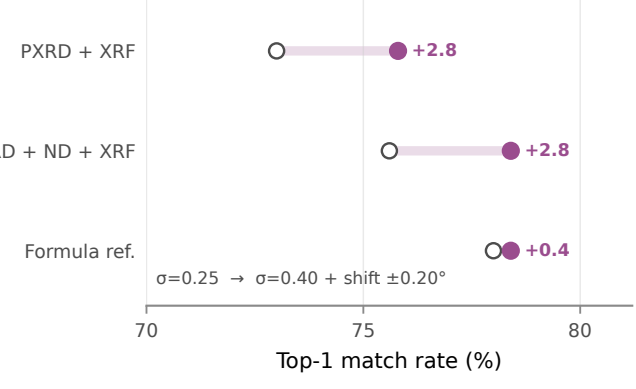
c Top-20 recall is more robust

Clean ○ / noisy ●; circle free, square known; N=500



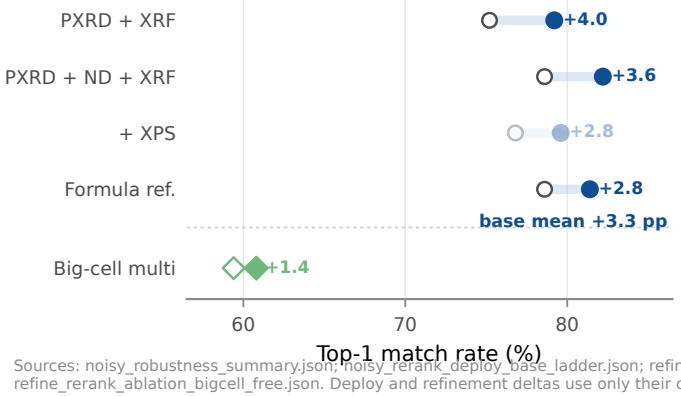
d Noisy-aware reranking

Same frozen candidates • noisy + lattice-free only



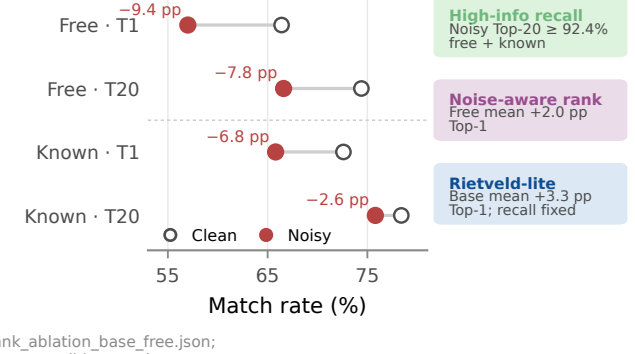
e Rietveld-lite reranking

Within each refinement pool • noisy + lattice-free • top-25



f Robust recall, residual ranking gap

Big-cell specialist; within-pool deltas only



Sources: noisy_robustness_summary.json; noisy_rerank_deploy_base_ladder.json; refine_rerank_ablation_base_free.json; refine_rerank_ablation_bigcell_free.json. Deploy and refinement deltas use only their own frozen candidate pools.